

Divya Rachala

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Professional Summary

Computer Science graduate with hands-on project experience in data analytics, machine learning, applied AI, and full-stack software development. Built reproducible Python and SQL pipelines, Power BI-ready KPI models, scikit-learn forecasting workflows, citation-grounded retrieval systems, and tested Java, Spring Boot, FastAPI, and Next.js applications. Seeking an entry-level role in data, machine learning, applied AI, or software engineering.

Technical Skills

Languages: Python, SQL, Java, TypeScript, JavaScript

Data and Business Intelligence: pandas, DuckDB, Parquet, Power Query, DAX, Power BI, Matplotlib

Machine Learning and AI: scikit-learn, regression, time-series validation, TF-IDF, information retrieval, natural language processing, ranking, model evaluation

Backend, Web, and Tools: Next.js, React, Spring Boot, FastAPI, REST APIs, Git, GitHub Actions, Vercel, pytest, JUnit, Vitest

Project Experience

Sunshine Retail Platform

Jun 2026 - Present

Full-Stack and Backend Project | github.com/divyarachala1812/sunshine-retail-platform

- Developed and deployed an Indian retail application using Next.js, TypeScript, Java 17, Spring Boot, Python, FastAPI, and Vercel; connected a searchable 50-product catalogue, cart, recommendations, checkout, order history, and customer support.
- Implemented inventory reservation, commit, and release rules across three checkout outcomes and eight delivery stages; verified frontend and backend behaviour with 26 Vitest, JUnit, and pytest tests.

KrishiGuide Scheme Assistant

Mar 2026 - May 2026

Applied AI Project | github.com/divyarachala1812/krishiguide-scheme-assistant

- Created a citation-grounded retrieval workflow over 406 chunks from seven official source items using word and character TF-IDF, hybrid ranking, Hindi and Hinglish query expansion, and extractive answer assembly.
- Designed an 18-question evaluation across five farmer-scheme topics; achieved Hit@1 of 1.000, expected-term coverage@3 of 0.972, and 100% citation completeness with pytest contracts and recorded source provenance.

AirWise Metro Forecasting

Aug 2025 - Dec 2025

Machine Learning Project | github.com/divyarachala1812/airwise-metro-forecasting

- Built a leakage-safe next-day PM2.5 forecasting pipeline with Python, pandas, and scikit-learn for Delhi, Mumbai, and Hyderabad; engineered lag, rolling, weather, and seasonal features from 3,747 city-day rows.
- Evaluated Ridge regression against persistence on 1,095 unseen 2025 rows, achieving MAE 8.50, R2 0.851, and alert recall 0.871; published city-level error analysis and permutation importance.

RidePulse Operations Analytics

Feb 2025 - May 2025

Data Analytics and Business Intelligence Project | github.com/divyarachala1812/ridepulse-operations-analytics

- Prepared and deduplicated 160,609 Delhi NCR ride records using Python, SQL, and DuckDB, producing 148,767 booking-level rows, 15 analytical marts, and governed KPI definitions.
- Designed a six-page Power BI implementation package with Power Query and DAX for demand, fulfilment, booking value, market, and service quality; passed 12 warehouse checks and three Python tests.

Education

Bachelor of Technology in Computer Science

Jun 2026

Malla Reddy Institute of Engineering and Technology, India